TULLY.

Autonomous workforce for fruit farms



Every orchard worked by an autonomous workforce

Orchard work will be automated. Tully builds the robots that make it reliable and affordable for growers, season after season, and sells the finished work, not the machines.

WHERE WE START

Apples
The largest labour pool of any tree fruit

Thinning, then harvesting
Thinning is simpler and reaches farmers sooner; harvesting uses the same machine

Europe's dense apple regions
South Tyrol and Bodensee first

WHERE WE GROW

More tasks
Pruning, and partner robots for spraying, mowing, scouting and bin logistics, all planned by one system

More tree fruit
Pears and other pome fruit, stone fruit, citrus, avocados

More regions
Hub by hub, and the Southern Hemisphere for a second season a year

WHAT WE LEAVE OUT

Berries, vegetables, greenhouses
Different crops, different machines

Selling machines
Farmers book the work; Tully runs the fleet

Founders who have built robots and companies before



Daniel Sattel

Co-founder & CEO

Second-time founder; scaled an IoT mobility company to seven-figure revenue

AWAKE Mobility · KONUX · dyson



Harsh Munshi

Co-founder & CTO

15+ years in computer vision, including perception for autonomous systems

KOPERNIKUS AUTOMOTIVE · GRAYMATICS



Raphael Joppi

Co-founder & CPO

Takes robots from concept to deployment; grew up in an apple-farming family

KINEXON · FRANKA EMIKA

ADVISOR



Dr. Ines Hanrahan

Executive Director of the Washington Tree Fruit Research Commission; ten years in tree-fruit robotics research and grower needs.

Fewer hands, higher costs

Every fresh tree fruit is still picked by hand, even cherries.

COST

60–65%

of an apple orchard's operating costs in the field is labour: harvest first, then pruning



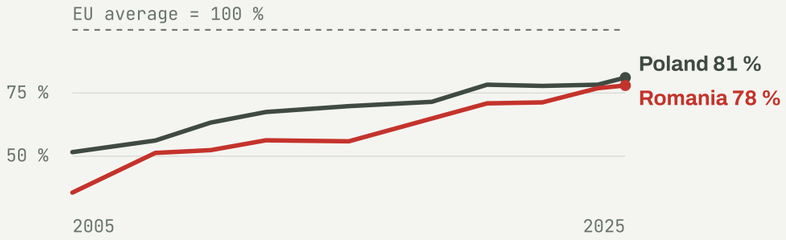
Spraying, fertiliser 25–35 % Other

Washington apples, 2024. Source: Gallardo, Galinato, Sallato-Carmona, Washington State University, 2025

DEMOGRAPHICS

−27%

seasonal workers on German farms, 2010 to 2023, as their home countries catch up



GDP per head in PPS, % of the EU average. Sources: Destatis farm census, 2010: 330,500 and 2023: 242,800 seasonal workers; Eurostat, 2025 flash estimate

> *"We need affordable and reliable automation, otherwise we will go out of business."*
>
> APPLE GROWER, POM CLUB MEETING



Fruit farm work, automated

An autonomous robot for the field work in orchards. One base, a kit for each job: thinning first, then harvesting.

Base + Arms = THINNING

Base + Arms + Fruit line, bin = HARVESTING

MANUAL WORK IN AN APPLE ORCHARD

703 h of hand work per hectare, each year

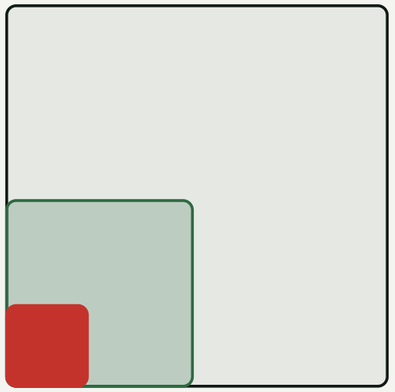
Harvesting	Thinning	Pruning	Other
375 h	105 h	100 h	123 h

Green: the work Tully takes on first. Other: fruit transport 33 h and other work 90 h. 5 ha family farm in South Tyrol. Source: South Tyrolean Extension Service for Fruit- and Winegrowing, Costi in melicoltura integrata, 2026 (Laimburg Research Centre)



TULLY H2 COBOT HARVESTER · RENDER

Orchards spend 60 bn € a year on manual work



TAM

60 bn €

manual work in orchards worldwide

SAM

14.3 bn €

apple harvesting and thinning worldwide

SOM

2.7 bn €

apple harvesting and thinning in 9 countries of western Europe and North America

WHY APPLES: THE MOST HAND WORK OF ANY TREE FRUIT



Other tree fruit: citrus, stone fruit, pears and other pome fruit, avocados. Sources: FAOSTAT, apple area 2022; Tully white paper. SAM and SOM: the harvesting and thinning share of apple hand work (68 %, Laimburg 2026). EU regions: USDA ERS

68%

of the hand work in an apple orchard is harvesting and thinning

≈8,700€

of hand labour per hectare a year in the obtainable market

462,000ha

of apple orchards in the obtainable market

75%

of EU apples grow in four countries, in dense regions

Farmers book the work, not the machine

THE FARMER BOOKS

- **One order**
  The blocks, the timing and the quality required
- **Paid per hectare**
  Compares directly with a seasonal crew
- **Nothing to buy**
  No machine, no upfront investment

TULLY DELIVERS

- **The work, done**
  Tully deploys, operates, supervises and maintains the robots
- **Responsible for the result**
  Quality of the work and the work done, against agreed requirements
- **A fleet busy all season**
  Thinning, then harvesting: more revenue than selling a robot once

PRICE PER HECTARE

THINNING

≈1,320 €

per hectare, against ≈1,470 € by hand: 105 h at 14 € an hour

HARVESTING

5,576 €

per hectare, against 6,195 € for a team of six on a picking platform

Both 10 % below the work by hand. Hours: Laimburg, 2026. Wage: South Tyrol farm workers' agreement, 12.11 € an hour gross, plus employer costs

RUN FROM REGIONAL HUBS

Orchards cluster
Climate and terroir pack fruit farms into dense regions

One hub per region
Storage, dispatch, maintenance, live fleet monitoring

Low cost to serve
Many customers close by: little overhead, fast help in the field

Each robot pays for itself in 1.3 seasons

ASSUMPTIONS · ONE ROBOT IN SOUTH TYROL

ROBOT	**260,000 €**, 8 off-the-shelf arms
SPEED	**0.7 s** per apple, whole machine; a team of six on a platform: 0.9 s
HOURS	**20 h** a day at 85 % uptime; a crew works 10 h
SEASON	**133 days**: 7 months of thinning and harvesting
AREA	**41 ha** a season, 3.3 days per hectare
PRICE	**5,576 €** per hectare, 10 % below a picking platform
RUNNING	**6,251 €** operation and **26,000 €** maintenance a season
LIFETIME	**10+ years**

Source: Tully white paper, unit economics, with a calculator to change the assumptions

ONE ROBOT

1 SEASON

195,658 €

gross profit, 86 % margin

Revenue	227,909 €
Operation	-6,251 €
Maintenance	-26,000 €

1.3 SEASONS

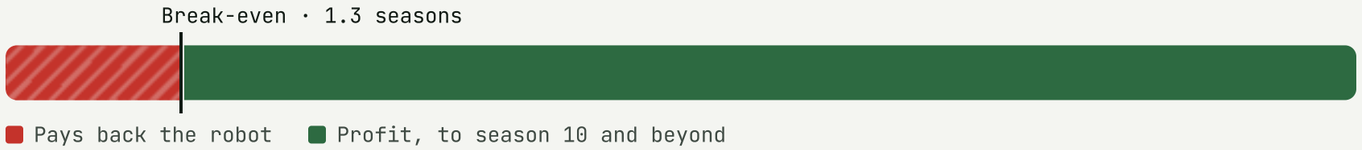
Break-even

The robot's 260,000 € cost is earned back

10 SEASONS

≈1.7 M€

gross profit after the robot's cost: 6.5 times what the robot cost



No apple robot has reached commercial use yet

APPLES	HOW IT PICKS	WHERE IT STANDS	WHAT GROWERS AND EXPERTS TELL US
TULLY.	Six-axis arms that reach around branches; cutting gripper; cameras on the base and in each hand	Simulation, perception pipeline	Built with growers: a co-founder from an apple-farming family, pilot orchards in South Tyrol
advanced.farm	Robotic arms with suction	Assets bought by CNH, 2025	Suction drops about 20 % of the fruit; very loud. CNH continues it, but quietly
abundant ROBOTICS	Vacuum picking in a dedicated machine	Closed, 2021	Did not work with the farming ecosystem; over-engineered and costly to build
TEVEL	Flying drones with suction, tethered to a platform	Field pilots	Good marketing; the market reports poor results in apple orchards
FF Robotics	Linear telescopic arms, mechanical grip	Field tested	High drop rate; a complex machine; linear arms get stuck on branches and leaves
AgriDynamics	Two-arm modules with soft vacuum grippers	Prototype, tests in Washington	Michigan State spin-off, 2025; reports 3–4 s per apple and 85 % picked; suction, like Advanced Farm
Finite Robotics	Thinning only: arms on a cart cut, suck and eject fruitlets	Field tests, 14 orchards, 2026	Grower-founder in Ontario, no venture capital; about 1,000 fruitlets an hour; 20 arms planned for 2027
Other fruits	Agrobot, Harvest CROO and Dogtooth in strawberries; Picker in citrus and avocados		

Sources: Michigan State University and Axios (AgriDynamics); Farmtario and FreshPlaza (Finite Robotics); Abundant Robotics closure; Tully white paper and Tully's interviews

WHY THEY STRUGGLED, AND WHAT TULLY DOES

~~Built from scratch~~ → Off-the-shelf arms, bases and cameras

~~Too long to market~~ → Thinning first, commercial in year 2

~~Suction, linear arms~~ → Six-axis arms that reach around branches and cut the stem

~~One view of the tree~~ → Cameras on the base and in each hand, AI models trained on our own field data

~~Far from growers~~ → A farming family in the team, pilots with growers from day one

Growers and experts want this machine

FROM FARMERS

~1,000 ha

of apple orchards whose growers have asked for Tully

30+

conversations with growers, cooperatives, experts and research centres

> *"Even if you can only automate one part of my manual work, the machine would run right now in my fields."*
>
> GROWER, INTERVIEWED BY TULLY

FROM EXPERTS AND INDUSTRY

> *"Automating the manual work would revolutionise the complete market."*
>
> INDUSTRY EXPERT, INTERVIEWED BY TULLY

- **Dr. Ines Hanrahan**
  Washington Tree Fruit Research Commission, advises Tully on picking and fruit-quality benchmarks
- **Frucotec**
  Full access to and support for its orchard platform, the base of the first machine
- **Several growers**
  Committed to testing Tully in their own orchards

WHAT FARMERS WOULD BUY

On demand

Available when the fruit is ready



0 €

No upfront investment

−10 %

Cheaper per hectare than picking by hand on a platform

0 h

No extra time to organise or supervise

~~Recruiting~~ ~~Housing~~ ~~Transport~~ ~~Paperwork~~

Commercial thinning in year 2

THIS ROUND

YEAR 1

Thinning prototype

Two machines thin tree after tree on their own in pilot orchards; a picking baseline at the Laimburg research centre

YEAR 2 COMMERCIAL

Commercial thinning

Thinning sold as a service in South Tyrol; first harvesting pilots with industrial apples

YEAR 3

Harvesting, regional hubs

Harvesting of top-grade apples; thinning run from hubs in South Tyrol, Bodensee and Washington

YEAR 4

New hubs

New hubs in Europe's dense apple regions; harvesting sold in every hub region

THEN

New fruits and regions

Pruning, pears and other tree fruit on the same base; the Southern Hemisphere

THE GOAL

What 100 M€ of revenue a year takes

440
robots

≈18,000 ha
harvested a year

≈50
people in regional field operations

≈30 %
of the orchards in South Tyrol and Germany alone

Buy what exists, build what matters

01

Moving through rows

- BUY Frucotec orchard platform
- BUY Nature Robots autonomy: RTK-GNSS and LiDAR

02

Perception

- BUY Colour and depth cameras on the base, a 3D camera in each hand, NVIDIA edge compute
- BUILD AI models for fruit, stems and branches
- BUILD Our own training data from the field and from packinghouse sorters

03

Manipulation

- BUY Eight six-axis cobot arms on vertical rails
- BUILD Cutting gripper with cameras in the fingertips
- BUILD Motion planning and visual servoing

04

Transport

- BUILD Soft fruit line from hand to bin, on-board sorter
- BUILD Self-driving bin units that swap full bins

Harvesting only, from year 3

HOW WE BUILD IT

Integrate

Off-the-shelf parts wherever they exist: low capital risk, fast to the field

Develop

AI, data generation and gripper design: where the robot wins

Make it robust

Iterate in the orchard until it runs all season with little supervision

NEXT STEP: THINNING

Off-season: base and autonomy tested in dormant orchards, arms on rails, thinning grippers, AI on mock fruit walls

After blossom: two machines thin in pilot orchards

3.5 M€ to prove the robot in the field

24 months and two growing seasons. Enough to run every work stream in parallel and be in the field as fast as possible. Details in the white paper.

TECH

- **Thinning prototype, then a robust version**
- **Harvesting prototype and gripper design**

COMMERCIAL

- **Commercial pilots of thinning as a service in South Tyrol**

USE OF FUNDS




SIMULATION · THE MACHINE AT WORK

TULLY.

Autonomous workforce for fruit farms

Raising 3.5 M€ over 24 months and two growing seasons to prove the robot in the field, then roll out region by region, hub by hub.



Daniel Sattel
Co-founder & CEO
daniel@tully-robotics.com

Harsh Munshi
Co-founder & CTO
harsh@tully-robotics.com

Raphael Joppi
Co-founder & CPO
raphael@tully-robotics.com